

Computer Networks



Overview of the Internet Protocol layering (Book 1 Ch 1)

Application Layer

- Application-Layer Paradigms
- Client-server applications
 - World Wide Web and HTTP
 - FTP
 - Electronic Mail
 - DNS
- Peer-to-peer paradigm
 - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

① Introduction

- Providing Services
- Application-Layer Paradigms

② Standard Client-Server Applications

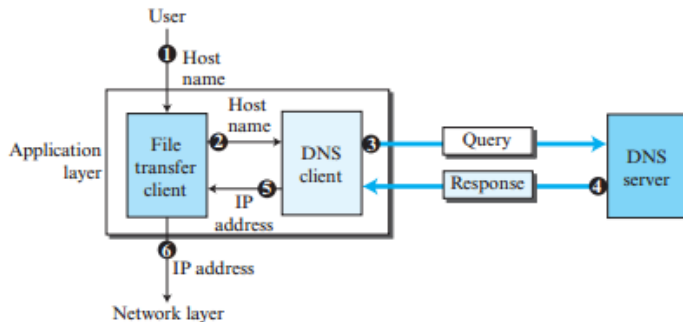
- World Wide Web and HTTP
- FTP
- Electronic Mail
- Domain Name System (DNS)

③ Peer-to-Peer Paradigm

- P2P Networks

Purpose of DNS

Figure 2.35 Purpose of DNS



Domain Name System (DNS)

- DNS stands for:
 - Domain Name System
- DNS designed to:
 - Help other application programs
- TCP/IP protocols use:
 - IP addresses for identification
- Humans prefer:
 - Names instead of numeric addresses
- DNS maps:
 - Host names to IP addresses
- Similar to:
 - Telephone directory service

Distributed Nature of DNS

- Internet is:
 - Extremely large
- Centralized directory system:
 - Not practical
 - Single failure can collapse system
- DNS uses:
 - Distributed database system
- DNS information distributed among:
 - Many computers worldwide
- Host contacts:
 - Nearest DNS server for mapping

DNS Name Resolution Process

- User knows:
 - Host name only
- Example:

afilesource.com

- TCP/IP requires:
 - IP address of remote host
- DNS client sends query to:
 - DNS server
- DNS server responds with:
 - Corresponding IP address
- File transfer client then:
 - Uses IP address to connect to server

Steps in DNS Mapping

- Step 1:
 - User gives host name to application client
- Step 2:
 - Client passes host name to DNS client
- Step 3:
 - DNS client sends query to DNS server
- Step 4:
 - DNS server returns IP address
- Step 5:
 - DNS client passes IP address to application client
- Step 6:
 - Application client accesses remote server

Connections in DNS Operation

- DNS operation requires:
 - Additional communication connection
- First connection:
 - DNS client ↔ DNS server
- Purpose:
 - Map host name to IP address
- Second connection:
 - Application client ↔ Remote server
- Purpose:
 - Actual service communication
- DNS mapping may sometimes require:
 - More than one DNS connection

DNS Name Space

- Machine names must be:
 - Carefully selected
 - Unique
- DNS uses:
 - Name space for mapping names to IP addresses
- Unique names required because:
 - IP addresses are unique
- Name spaces can be:
 - Flat
 - Hierarchical

Flat Name Space

- In flat name space:
 - One name assigned to one address
- Name characteristics:
 - Sequence of characters
 - No internal structure
- Common sections in names:
 - Have no meaning
- Main disadvantage:
 - Requires centralized control
- Not suitable for:
 - Large systems like the Internet
- Reason:
 - Difficult to avoid duplication and ambiguity

Hierarchical Name Space

- Hierarchical name space:
 - Name divided into several parts
- Different parts can define:
 - Nature of organization
 - Organization name
 - Department
 - Host or resource
- Advantages:
 - Decentralized management
 - Easier administration
- Central authority assigns:
 - Main organization name
- Organization itself assigns:
 - Remaining suffixes or prefixes



Example of Hierarchical Naming

- Two organizations may use:
 - Same host name
- Example host name:

caesar

- Organization 1 domain:

first.com

- Organization 2 domain:

second.com

- Final unique names:

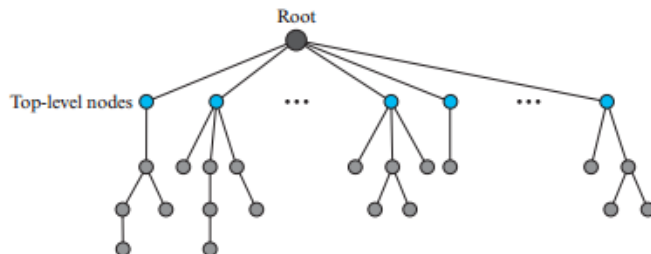
caesar.first.com

caesar.second.com

- Result:
 - Names remain unique

Domain name space

Figure 2.36 *Domain name space*



Domain Name Space

- Hierarchical naming implemented using:
 - Domain name space
- Domain name space uses:
 - Inverted-tree structure
- Tree characteristics:
 - Root at the top
 - Maximum 128 levels
- Levels include:
 - Level 0 = Root
 - Level 127 = Lowest level

Labels in Domain Name Space

- Each node in tree has:
 - Label
- Label:
 - String with maximum 63 characters
- Root label:
 - Null string (empty string)
- DNS rule:
 - Children of same node must have different labels
- Result:
 - Guarantees unique domain names

Domain Names

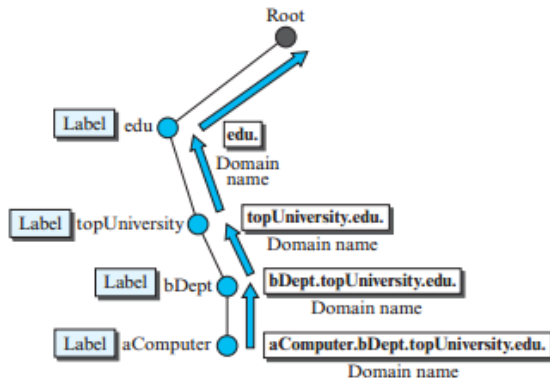
- Each node has:
 - Domain name
- Full domain name:
 - Sequence of labels separated by dots (.)
- Domain names read:
 - From node toward root
- Last label:
 - Root label (null string)
- Therefore:
 - Full domain name ends with dot (.)

FQDN and PQDN

- FQDN:
 - Fully Qualified Domain Name
- Characteristics of FQDN:
 - Ends with null label
 - Ends with dot (.)
- PQDN:
 - Partially Qualified Domain Name
- Characteristics of PQDN:
 - Does not reach root
 - Missing suffix part
- PQDN used when:
 - Client and target belong to same site
- Resolver adds:
 - Missing suffix to create FQDN

Domain name and Labels

Figure 2.37 *Domain names and labels*

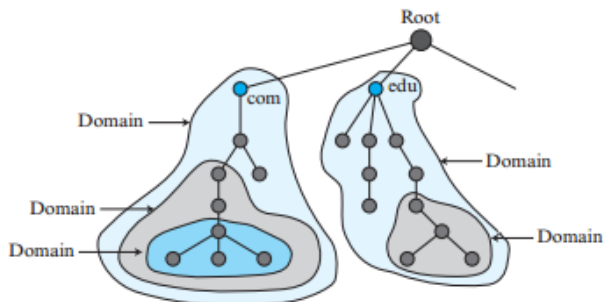


Domain

- A domain is:
 - Subtree of the domain name space
- Domain name:
 - Name of node at top of subtree
- Domain structure:
 - Hierarchical
- A domain can contain:
 - Subdomains
- Domain division allows:
 - Easier management
 - Decentralized administration
- Example:
 - com domain can contain:
 - google.com
 - yahoo.com

Domains

Figure 2.38 Domains



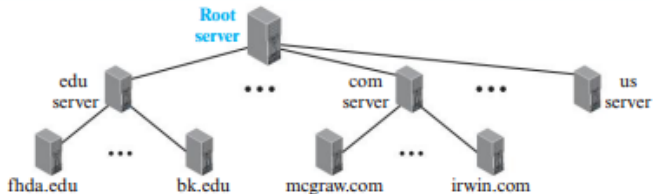
Distribution of Name Space

- Domain name space contains:
 - Huge amount of information
- Information must be:
 - Stored efficiently
- Storing all information on one computer:
 - Inefficient
 - Unreliable
- Problem with centralized storage:
 - Heavy load from worldwide requests
- Centralized system failure may:
 - Disrupt entire DNS operation
- DNS solves problem using:
 - Distributed storage of name space



Hierarchy of name servers

Figure 2.39 *Hierarchy of name servers*



Hierarchy of Name Servers

- DNS information distributed among:
 - Multiple DNS servers
- Distribution method:
 - Divide name space into domains
- Root node:
 - Remains separate
- Domains created based on:
 - First-level nodes
- Example:
 - com
 - edu
 - org

Subdomains and DNS Authority

- Large domains can be:
 - Further divided into subdomains
- Each DNS server responsible for:
 - One domain or subdomain
- Server responsible for domain called:
 - Authoritative server
- DNS supports:
 - Large domains
 - Small domains
- Result:
 - Hierarchy of DNS servers
- DNS hierarchy matches:
 - Hierarchy of domain names



Zone in DNS

- Entire domain name hierarchy:
 - Cannot be stored on one server
- DNS hierarchy divided among:
 - Multiple DNS servers
- Portion of hierarchy managed by a server:
 - Called a zone
- Zone defined as:
 - Contiguous part of DNS tree

Domain and Zone Relationship

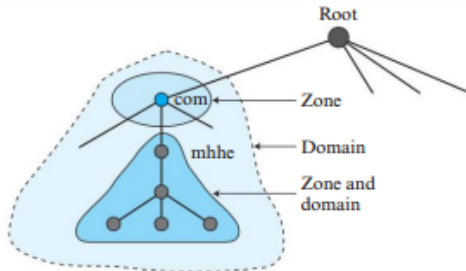
- If domain is not divided:
 - Domain and zone are same
- Server stores:
 - Complete information about all nodes
- Server creates:
 - Zone file (database)
- Zone file contains:
 - Information for every node in domain

Delegation of Authority

- Large domains may be:
 - Divided into subdomains
- Authority delegated to:
 - Lower-level DNS servers
- In this case:
 - Domain and zone are different
- Detailed subdomain information stored at:
 - Lower-level servers
- Original server keeps:
 - References to lower-level servers
- Original server still:
 - Maintains partial authority over zone

Zone

Figure 2.40 Zone



Root Server

- Root server:
 - Server whose zone consists of whole DNS tree
- Root server usually:
 - Does not store detailed domain information
- Root server mainly:
 - Delegates authority to other servers
- Root server keeps:
 - References to lower-level servers
- Multiple root servers exist:
 - Distributed around the world
- Each root server covers:
 - Entire domain name space

Primary Server

- Primary server:
 - Stores original zone file
- Responsible for:
 - Creating zone file
 - Maintaining zone file
 - Updating zone file
- Zone file stored on:
 - Local disk
- Primary server is:
 - Authoritative for its zone
- Primary server loads information:
 - Directly from disk file

Secondary Server

- Secondary server:
 - Receives zone information from another server
- Source of information:
 - Primary server
 - Another secondary server
- Secondary server:
 - Does not create zone files
 - Does not update zone files
- Updates performed by:
 - Primary server only
- Secondary server stores:
 - Copied zone file on local disk
- Secondary server loads information:
 - From primary server

Primary vs Secondary Servers

- Both primary and secondary servers are:
 - Authoritative servers
- Purpose of secondary server:
 - Provide redundancy
 - Improve reliability
- If one server fails:
 - Other server continues service
- A DNS server can act as:
 - Primary server for one zone
 - Secondary server for another zone
- Server role depends on:
 - Specific zone being discussed

DNS in the Internet

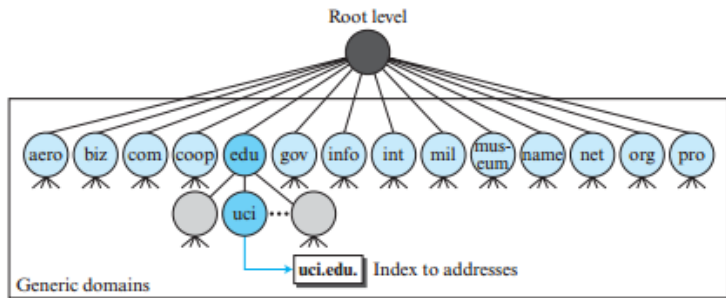
- DNS:
 - Protocol usable on different platforms
- Internet domain name space originally divided into:
 - Generic domains
 - Country domains
 - Inverse domains
- Inverse domains: 
 - Used to map IP address to host name
- Due to Internet growth:
 - Inverse domains became difficult to manage
- Inverse domains are now:
 - Deprecated (RFC 3425)
- Current focus:
 - Generic domains
 - Country domains

Generic Domains

- Generic domains classify hosts based on:
 - Generic behavior or function
- Each node in DNS tree:
 - Defines a domain
- Domain acts as:
 - Index to DNS database
- Examples of generic domains:
 - com — Commercial organizations
 - edu — Educational institutions
 - org — Organizations
 - gov — Government agencies
- Generic domains simplify:
 - Organization and lookup of Internet hosts

Generic domains

Figure 2.41 Generic domains



First-Level Generic Domain Labels

- Generic domain tree:
 - Organized hierarchically
- First level of generic domains:
 - Allows 14 possible labels
- Each label represents:
 - Specific organization type
- Purpose of labels:
 - Classify hosts based on organization category
- Examples of labels:
 - com — Commercial organizations
 - edu — Educational institutions
 - gov — Government organizations
 - org — Nonprofit organizations
 - net — Network support organizations

Generic domain Labels

Table 2.12 *Generic domain labels*

<i>Label</i>	<i>Description</i>	<i>Label</i>	<i>Description</i>
aero	Airlines and aerospace	int	International organizations
biz	Businesses or firms	mil	Military groups
com	Commercial organizations	museum	Museums
coop	Cooperative organizations	name	Personal names (individuals)
<i>Label</i>	<i>Description</i>	<i>Label</i>	<i>Description</i>
edu	Educational institutions	net	Network support centers
gov	Government institutions	org	Nonprofit organizations
info	Information service providers	pro	Professional organizations

Country Domains

- Country domains use:
 - Two-character country abbreviations
- Example:
 - us — United States
- Country domains identify:
 - Geographic regions or countries
- Second-level labels may represent:
 - Organizations
 - National subdivisions

Example of Country Domain Structure

- United States uses:
 - State abbreviations as subdivisions
- Example:

ca.us

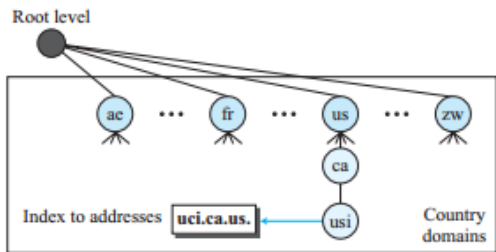
- Example domain:



- Interpretation:
 - University of California, Irvine
 - Located in California
 - Located in United States
- Country domains provide:
 - Geographic organization of DNS hierarchy

Country domains

Figure 2.42 Country domains



Resolution in DNS

- Mapping a name to an address:
 - Called name-address resolution
- DNS works as:
 - Client-server application
- Host uses:
 - DNS client called resolver
- Resolver sends request to:
 - Closest DNS server
- DNS server may:
 - Provide answer directly
 - Refer resolver to another server
 - Query other servers
- Resolution types:
 - Recursive resolution
 - Iterative resolution

Recursive Resolution Process

- Source host:

`some.anet.com`

- Destination host:

`engineering.mcgraw-hill.com`

- Application requests:
 - IP address of destination host
- Resolver sends query to:
 - Local DNS server at Anet ISP
- Local DNS server sends query to:
 - Root DNS server
- Root server forwards query to:
 - Top-level domain server for com

Recursive Resolution: Final Steps

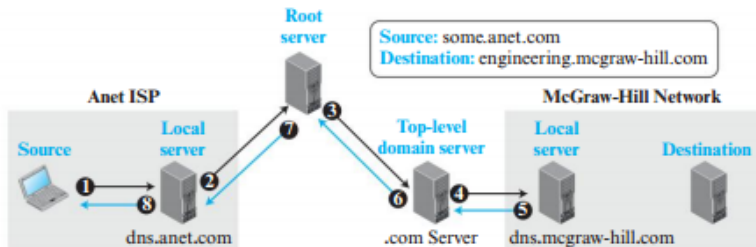
- Top-level domain server knows:
 - DNS server for McGraw-Hill domain
- Query sent to:

`dns.mcgraw-hill.com`

- McGraw-Hill DNS server:
 - Returns destination IP address
- IP address returned through:
 - Top-level DNS server
 - Root server
 - ISP DNS server
- ISP DNS server may:
 - Cache result for future use
- Resolver finally delivers:
 - IP address to application program

Recursive resolution

Figure 2.43 Recursive resolution



Iterative Resolution

- In iterative resolution:
 - Server does not fully resolve query
- If server does not know mapping:
 - Sends IP address of next DNS server
- Requesting server:
 - Continues querying other servers
- Final answer usually returned to:
 - Local DNS server
- Resolver receives:
 - Final resolved address from local DNS server

Iterative Resolution Process

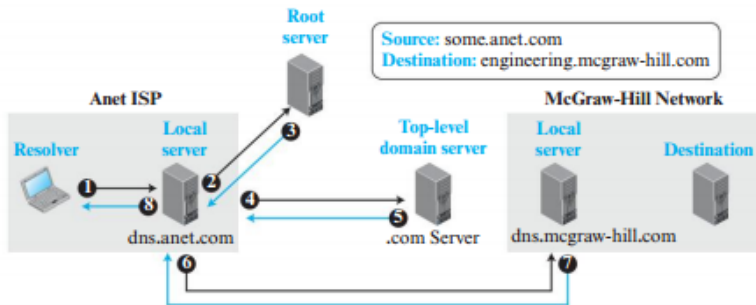
- Resolver sends query to:
 - Local DNS server
- Local DNS server queries:
 - Root DNS server
- Root server returns:
 - IP address of top-level domain server
- Local DNS server then queries:
 - Top-level domain server
- Top-level domain server returns:
 - IP address of McGraw-Hill DNS server

Final Steps in Iterative Resolution

- Local DNS server queries:
 - McGraw-Hill local DNS server
- McGraw-Hill DNS server returns:
 - IP address of destination host
- Messages in events:
 - 2, 4, and 6 contain same query
- Messages in events:
 - 3 returns top-level DNS server address
 - 5 returns McGraw-Hill DNS server address
 - 7 returns destination host address
- Local DNS server finally sends:
 - Destination IP address to resolver

Iterative resolution

Figure 2.44 Iterative resolution



Caching in DNS

- DNS servers may receive:
 - Queries for domains outside their authority
- Searching repeatedly for mappings:
 - Increases delay
- DNS improves efficiency using:
 - Caching
- Cache stores:
 - Previously resolved mappings
- Cached information used for:
 - Future similar queries

Working of DNS Caching

- Server requests mapping from:
 - Another DNS server
- After receiving response:
 - Mapping stored in cache memory
- Future queries for same mapping:
 - Resolved directly from cache
- Cached response marked as:
 - Unauthoritative
- Meaning:
 - Response came from cache
 - Not directly from authoritative server

TTL and Cache Management

- Problem with caching:
 - Cached mapping may become outdated
- Solution:
 - Time To Live (TTL)
- TTL:
 - Time in seconds for valid caching
- Authoritative server:
 - Adds TTL value to mapping
- After TTL expires:
 - Mapping becomes invalid
- DNS servers maintain:
 - TTL counter for cached mappings
- Expired mappings:
 - Removed from cache memory

Resource Records in DNS

- Zone information stored as:
 - Resource Records (RRs)
- DNS server maintains:
 - Database of resource records
- Resource record:
 - 5-tuple structure

(Domain Name, Type, Class, TTL, Value)

Fields of Resource Record

- Domain Name:
 - Identifies the resource record
- Type:
 - Defines interpretation of value field
- Class:
 - Defines network type
- Common class:
 - IN (Internet)
- TTL:
 - Validity time of information in seconds
- Value:
 - Stores actual information related to domain

Common Resource Record Types

- Type field determines:
 - Meaning of stored value
- Common RR types include:
 - A — Maps host name to IP address
 - NS — Defines authoritative name server
 - MX — Defines mail exchanger
 - CNAME — Alias for canonical name
 - PTR — Pointer to another part of domain space
- Resource records help DNS:
 - Resolve names efficiently
 - Store different types of information

Table 2.13 *Types*

<i>Type</i>	<i>Interpretation of value</i>
A	A 32-bit IPv4 address (see Chapter 4)
NS	Identifies the authoritative servers for a zone
CNAME	Defines an alias for the official name of a host
SOA	Marks the beginning of a zone
MX	Redirects mail to a mail server
AAAA	An IPv6 address (see Chapter 4)

DNS Messages

- DNS uses two types of messages:
 - Query message
 - Response message
- Both message types:
 - Use same message format
- DNS messages contain:
 - Header section
 - Question section
 - Answer section
 - Authoritative section
 - Additional information section

DNS Header Fields

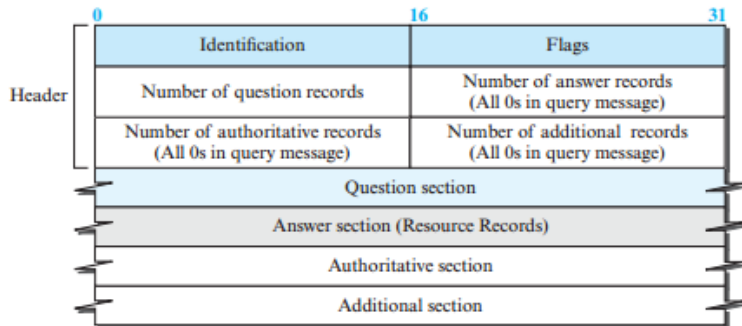
- Identification field:
 - Matches response with query
- Flag field:
 - Identifies query or response
 - Indicates error status
- Header also contains:
 - Number of records in each section
- Record count fields specify:
 - Number of questions
 - Number of answers
 - Number of authoritative records
 - Number of additional records

Sections of DNS Message

- Question section:
 - Contains one or more question records
 - Present in both query and response
- Answer section:
 - Contains resource records
 - Present only in response messages
- Authoritative section:
 - Provides information about authoritative servers
- Additional information section:
 - Supplies extra helpful information
- DNS message structure helps:
 - Efficient name resolution

DNS message

Figure 2.45 DNS message



Note:

The query message contains only the question section.
The response message includes the question section,
the answer section, and possibly two other sections.

Example: nslookup Utility

- nslookup utility:
 - Used for DNS name-address mapping
- Available in:
 - UNIX
 - Windows
- Purpose:
 - Retrieve IP address from domain name
- Example command:



```
$ nslookup www.forouzan.biz
```

- Output:

```
Name:  www.forouzan.biz
```

```
Address:  198.170.240.179
```

- Result:
 - Domain name resolved to IP address

DNS Encapsulation

- DNS can use:
 - UDP
 - TCP
- Well-known DNS server port:



- UDP used when:
 - Response size is less than 512 bytes
- Reason:
 - Most UDP packets limited to 512 bytes

Use of TCP in DNS

- TCP used when:
 - Response size exceeds 512 bytes
- Scenario 1:
 - Resolver already knows response is large
- Example:
 - Zone transfer from primary server to secondary server
- In this case:
 - Resolver directly uses TCP connection

UDP Truncation and TCP Retry

- Scenario 2:
 - Resolver does not know response size
- Resolver initially uses:
 - UDP
- If response exceeds 512 bytes:
 - Server truncates response
- Server then:
 - Sets TC (Truncated) bit
- Resolver actions:
 - Opens TCP connection
 - Repeats request
- Result:
 - Full response received using TCP

Registrars in DNS

- New domains added through:
 - Registrars
- Registrar:
 - Commercial entity accredited by ICANN
- Registrar responsibilities:
 - Verify domain name uniqueness
 - Enter domain into DNS database
- Domain registration:
 - Requires payment of fee
- Many registrars available:
 - Example website:

`http://www.internic.net`

Example of Domain Registration

- Organization example:

wonderful

- Server name:

ws

- IP address:



- Information submitted to registrar:

Domain name: ws.wonderful.com

IP address: 200.200.200.5

- Registrar then:
 - Adds domain entry to DNS database

Dynamic DNS (DDNS)

- Original DNS design:
 - Assumed fewer address changes
- Changes in DNS may include:
 - Adding new host
 - Removing host
 - Changing IP address
- Traditional DNS updates:
 - Required manual modification of DNS master file
- Modern Internet size:
 - Makes manual updates impractical
- Solution:
 - Dynamic Domain Name System (DDNS)

Working of DDNS

- DDNS automatically updates:
 - DNS records dynamically
- When name-address binding changes:
 - Information sent to primary DNS server
- Information usually sent by:
 - DHCP
- Primary server:
 - Updates the zone file
- Secondary servers are then:
 - Notified about changes

Notification and Zone Transfer

- Two notification methods:
 - Active notification
 - Passive notification
- Active notification:
 - Primary server informs secondary servers
- Passive notification:
 - Secondary servers periodically check for updates
- After notification:
 - Secondary server requests updated zone information
- Complete zone update process called:
 - Zone transfer
- DDNS security:
 - Uses authentication mechanism
 - Prevents unauthorized DNS changes

Security of DNS

- DNS is:
 - Critical part of Internet infrastructure
- Internet applications dependent on DNS:
 - Web access
 - E-mail
- DNS vulnerable to:
 - Multiple security attacks

Types of DNS Attacks

- Attack 1:
 - Attacker reads DNS responses
- Purpose:
 - Identify frequently accessed sites
 - Create user profile
- Prevention:
 - DNS message confidentiality
- Attack 2:
 - Attacker modifies DNS responses
 - Creates bogus DNS responses
- Purpose:
 - Redirect user to malicious site
- Prevention:
 - Message origin authentication
 - Message integrity



DNSSEC and Protection Mechanisms

- Attack 3:
 - Flooding DNS server
- Result:
 - Server overload
 - Denial-of-Service (DoS)
- IETF developed:
 - DNS Security (DNSSEC)
- DNSSEC provides:
 - Message origin authentication
 - Message integrity
- DNSSEC uses:
 - Digital signatures
- DNSSEC limitations:
 - No confidentiality protection
 - No direct DoS protection
- DNS caching helps:
 - Reduce impact of DoS attacks